

Predicting UEFA Champions League Match Outcomes Using Machine Learning

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Abstract—This project explores how machine learning can predict the outcomes of UEFA Champions League matches using team and player performance data. Using a data set of match statistics such as goals, expected goals (xG), and possession, three models—Random Forest, Logistic Regression, and XGBoost—were trained and compared. The results show that ensemble models like Random Forest and XGBoost perform better at capturing patterns in the data. The study highlights which features, such as Matches Played, Points, and Goal Difference, most influence match outcomes.

I. INTRODUCTION

Traditional approaches that rely on simple standings, win-loss trends, or single summary statistics struggle to capture nonlinear patterns and interactions in modern football data (e.g., schedule strength, recent form, and variance). Machine learning can model these relationships more flexibly, which motivates comparing multiple algorithms on the same dataset. The UEFA Champions League is one of the most competitive tournaments in the world. Predicting match outcomes is both challenging and valuable for coaches, fans, and analysts. The goal of this project is to explore how machine learning can forecast match results using team and player statistics. With advances in AI, sports analytics has become an exciting field that uses data to find patterns and improve predictions.

II. LITERATURE REVIEW

Recent studies have shown how different algorithms can improve the accuracy of sports prediction. Wang et al. (2025) used a transformer model to study how players and teams interact on the field. This helps capture more realistic match patterns. Wang et al., 2025. Proposes HIGFormer, a transformer that models player-team interactions to predict match outcomes. On benchmark soccer data, interaction modeling improves accuracy over classical ML. Shows value of structured, context-aware features.

Hewitt Karakuş, 2023. Introduces position-adjusted xG, correcting expected-goals by player role and context. The adjusted features improve predictive performance vs. raw xG, suggesting feature engineering matters as much as model choice.

Divekar et al., 2023. Uses Bayesian updating for in-match forecasting with calibrated uncertainty. Produces real-time win-probability curves and credible intervals—strength is probabilistic interpretability.

Bunker et al., 2024. Survey comparing ML methods for soccer prediction (logistic, trees, ensembles, DL) and metrics (accuracy/F1/AUC). Concludes ensemble methods are reliable baselines and stresses data quality and evaluation design.

Kinalioglu et al., 2021. Applies classical classifiers (SVM, RF, LR) to Champions League knockout ties. Finds Random Forest frequently outperforms linear models, highlighting non-linear relations in tournament data.

Ensembles (Random Forest/XGBoost) generally outperform linear models across studies (Bunker; Kinalioglu), while Wang shows transformers can add value when rich interaction data exist. Hewitt Karakuş emphasize feature quality (position-adjusted xG) as a key driver regardless of model. Divekar focuses on probabilistic, real-time forecasts rather than pre-match classification, trading structure for calibrated uncertainty.

Most work either (i) evaluates a single model family, (ii) relies on player-level or proprietary tracking data, or (iii) targets older seasons or different tasks (e.g., in-match probabilities). There are fewer reproducible comparisons of multiple models on recent Champions League, team-level features with clear interpretability. Our study addresses this by testing LR, RF, and XGBoost on the same 2023–24 dataset and reporting both performance and feature importance. Open questions include: how results change with multi-season data, richer features (possession/pressing), and uncertainty estimates. Together, these papers show that the use of advanced models and the right performance metrics can make soccer prediction more reliable. While prior studies improved soccer prediction in different ways, many focus on single model families, single-season case studies, or player-centric pipelines that are costly to reproduce. Fewer works provide a simple, reproducible comparison of multiple models on recent Champions League team-level data with clear interpretability. In this project, I address this gap by training and evaluating three widely used models on the same dataset, reporting performance, and visualizing which features drive predictions.

III. METHODOLOGY

I used Kaggle data sets that included match data, team stats, and performance metrics. The project was done in Google Colab using Python and libraries such as Pandas, Matplotlib, Seaborn, and Scikit-Learn. I trained and tested three models: Logistic Regression, Random Forest, and XGBoost. The data set was divided by 75% for training and 25% for testing. The models were evaluated using the results of the accuracy, precision, recall, and confusion matrix.

IV. RESULTS

The Random Forest and XGBoost models gave the best accuracy. Matches Played (MP), Points (Pts), and Goal Differ-

ence (GD) were the most important features. Expected Goals (xG) and Expected Goal Difference (xGD) also influenced predictions.

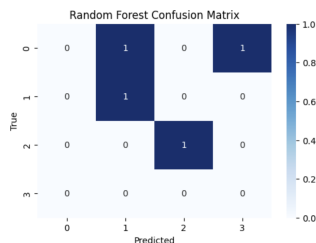


Fig. 1. Confusion matrix showing how well the Random Forest model predicted each tournament stage.

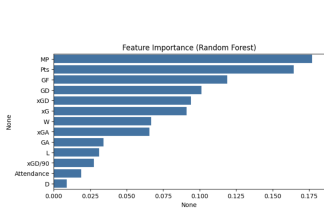


Fig. 2. Feature importance showing which match statistics most influenced predictions.

V. DISCUSSION AND CONCLUSION

The models performed fairly well, even though the dataset was small. Random Forest and XGBoost captured more complex patterns than Logistic Regression. The results showed that team consistency (MP and Pts) and scoring balance (GD and xG) are key to success. If this project were continued, I would use more seasons of data and include player-level stats to improve accuracy and explore deep learning models for better predictions.

REFERENCES

REFERENCES

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